

studentai

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Chapter 1

Hierarchical Index

1.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Human	7
Studentas	8

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Human		7
Studentas	Represents a student with homework grades and an exam score, providing calculations for mean- and median-based final grades	8

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

Include/ human.h	15
Include/ my_functions.h	15
Include/ my_library.h	22
Include/ student.h	
Declaration of the Studentas class, extending Human to manage grades and compute final scores	23
src/ additional_functions.cpp	24
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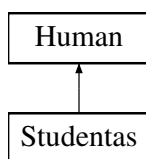
Chapter 4

Class Documentation

4.1 Human Class Reference

```
#include <human.h>
```

Inheritance diagram for Human:



Public Member Functions

- `Human` (const string &`vardas`, const string &`pavarde`)
- string `getVardas` () const
- string `getPavarde` () const
- virtual void `unusedFunction` () const =0
- virtual `~Human` ()

Protected Attributes

- string `vardas`
- string `pavarde`

4.1.1 Constructor & Destructor Documentation

4.1.1.1 Human()

```
Human::Human (  
    const string & vardas,  
    const string & pavarde) [inline]
```

4.1.1.2 ~Human()

```
virtual Human::~~Human () [inline], [virtual]
```

4.1.2 Member Function Documentation

4.1.2.1 getPavarde()

```
string Human::getPavarde () const [inline]
```

4.1.2.2 getVardas()

```
string Human::getVardas () const [inline]
```

4.1.2.3 unusedFunction()

```
virtual void Human::unusedFunction () const [pure virtual]
```

Implemented in [Studentas](#).

4.1.3 Member Data Documentation

4.1.3.1 pavarde

```
string Human::pavarde [protected]
```

4.1.3.2 vardas

```
string Human::vardas [protected]
```

The documentation for this class was generated from the following file:

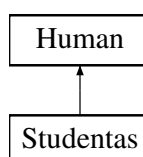
- Include/[human.h](#)

4.2 Studentas Class Reference

Represents a student with homework grades and an exam score, providing calculations for mean- and median-based final grades.

```
#include <student.h>
```

Inheritance diagram for Studentas:



Public Member Functions

- [Studentas](#) ()
Default constructor.
- [Studentas](#) (const std::string &[vardas](#), const std::string &[pavarde](#), const std::vector< float > &pazymiai, const int &egzaminoRezultatas)
Parameterized constructor. Initializes student name, surname, grades vector, and exam result, then computes final grades.
- [~Studentas](#) ()
Destructor.
- [Studentas](#) (const [Studentas](#) &other)
Copy constructor.
- [Studentas](#) ([Studentas](#) &&other) noexcept
Move constructor.
- [Studentas](#) & [operator=](#) (const [Studentas](#) &other)
Copy assignment operator.
- [Studentas](#) & [operator=](#) ([Studentas](#) &&other) noexcept
Move assignment operator.
- void [setVardas](#) (const std::string &v)
- void [setPavarde](#) (const std::string &p)
- void [setGrades](#) (const std::vector< float > &g)
- void [setExam](#) (const int &exam)
- std::vector< float > [getPazymiai](#) () const
- int [getEgzaminoRezultatas](#) () const
- float [getGalutinisVid](#) () const
- float [getGalutinisMed](#) () const
- void [unusedFunction](#) () const override
Implementation of unused pure virtual from [Human](#).
- float [calculateMean](#) ()
Calculate mean of homework grades.
- float [calculateMedian](#) ()
Calculate median of homework grades.
- void [calculate_everything](#) ()
Compute both mean- and median-based final grades.

Public Member Functions inherited from [Human](#)

- [Human](#) (const string &[vardas](#), const string &[pavarde](#))
- string [getVardas](#) () const
- string [getPavarde](#) () const
- virtual [~Human](#) ()

Static Public Member Functions

- static void [appendingContainerViaFile](#) (const std::string &file_name, std::vector< [Studentas](#) > &student_list)
Read students from a text file and append them to a container.
- static std::vector< [Studentas](#) > [read_student_records](#) (int ndCount, std::istream &iss)
Parse student records from a string stream.
- static void [print_to_file](#) (const std::vector< [Studentas](#) > &list_of_students, const std::string &file_name)
Write a list of students to a text file under "Rezultatai" folder.

Friends

- `std::ostream & operator<<` (`std::ostream &out`, `const Studentas &s`)
Print surname, name and both final grades (mean- and median-based).
- `std::istream & operator>>` (`std::istream &in`, `Studentas &s`)
Prompt for surname, name, homework grades, and exam score; then compute final grades.

Additional Inherited Members

Protected Attributes inherited from [Human](#)

- `string vardas`
- `string pavarde`

4.2.1 Detailed Description

Represents a student with homework grades and an exam score, providing calculations for mean- and median-based final grades.

4.2.2 Constructor & Destructor Documentation

4.2.2.1 Studentas() [1/4]

```
Studentas::Studentas () [inline]
```

Default constructor.

4.2.2.2 Studentas() [2/4]

```
Studentas::Studentas (
    const std::string & vardas,
    const std::string & pavarde,
    const std::vector< float > & pazymiai,
    const int & egzaminoRezultatas) [inline]
```

Parameterized constructor. Initializes student name, surname, grades vector, and exam result, then computes final grades.

Parameters

<i>vardas</i>	First name of the student.
<i>pavarde</i>	Surname of the student.
<i>pazymiai</i>	Vector of homework grades.
<i>egzaminoRezultatas</i>	Integer score of the exam.

4.2.2.3 ~Studentas()

```
Studentas::~~Studentas () [inline]
```

Destructor.

4.2.2.4 Studentas() [3/4]

```
Studentas::Studentas (
    const Studentas & other)
```

Copy constructor.

4.2.2.5 Studentas() [4/4]

```
Studentas::Studentas (
    Studentas && other) [noexcept]
```

Move constructor.

4.2.3 Member Function Documentation

4.2.3.1 appendingContainerViaFile()

```
void Studentas::appendingContainerViaFile (
    const std::string & file_name,
    std::vector< Studentas > & student_list) [static]
```

Read students from a text file and append them to a container.

Parameters

<i>file_name</i>	Path to the input file.
<i>student_list</i>	Container (vector) to which parsed <code>Studentas</code> objects are appended.

4.2.3.2 calculate_everything()

```
void Studentas::calculate_everything ()
```

Compute both mean- and median-based final grades.

4.2.3.3 calculateMean()

```
float Studentas::calculateMean ()
```

Calculate mean of homework grades.

4.2.3.4 calculateMedian()

```
float Studentas::calculateMedian ()
```

Calculate median of homework grades.

4.2.3.5 getEgzaminoRezultatas()

```
int Studentas::getEgzaminoRezultatas () const [inline]
```

4.2.3.6 getGalutinisMed()

```
float Studentas::getGalutinisMed () const [inline]
```

4.2.3.7 getGalutinisVid()

```
float Studentas::getGalutinisVid () const [inline]
```

4.2.3.8 getPazymiai()

```
std::vector< float > Studentas::getPazymiai () const [inline]
```

4.2.3.9 operator=() [1/2]

```
Studentas & Studentas::operator= (  
    const Studentas & other)
```

Copy assignment operator.

4.2.3.10 operator=() [2/2]

```
Studentas & Studentas::operator= (  
    Studentas && other) [noexcept]
```

Move assignment operator.

4.2.3.11 print_to_file()

```
void Studentas::print_to_file (  
    const std::vector< Studentas > & list_of_students,  
    const std::string & file_name) [static]
```

Write a list of students to a text file under "Rezultatai" folder.

Parameters

<i>list_of_students</i>	Container of Studentas objects to write.
<i>file_name</i>	Name of the output file.

4.2.3.12 read_student_records()

```
vector< Studentas > Studentas::read_student_records (
    int ndCount,
    std::istream & iss) [static]
```

Parse student records from a string stream.

Parameters

<i>ndCount</i>	Number of homework grades per record.
<i>iss</i>	Input string stream containing student data.

Returns

Vector of parsed [Studentas](#) objects.

4.2.3.13 setExam()

```
void Studentas::setExam (
    const int & exam) [inline]
```

4.2.3.14 setGrades()

```
void Studentas::setGrades (
    const std::vector< float > & g) [inline]
```

4.2.3.15 setPavarde()

```
void Studentas::setPavarde (
    const std::string & p) [inline]
```

4.2.3.16 setVardas()

```
void Studentas::setVardas (
    const std::string & v) [inline]
```

4.2.3.17 unusedFunction()

```
void Studentas::unusedFunction () const [override], [virtual]
```

Implementation of unused pure virtual from [Human](#).

Implements [Human](#).

4.2.4 Friends And Related Symbol Documentation

4.2.4.1 operator<<

```
std::ostream & operator<< (  
    std::ostream & out,  
    const Studentas & s) [friend]
```

Print surname, name and both final grades (mean- and median-based).

4.2.4.2 operator>>

```
std::istream & operator>> (  
    std::istream & in,  
    Studentas & s) [friend]
```

Prompt for surname, name, homework grades, and exam score; then compute final grades.

The documentation for this class was generated from the following files:

- Include/[student.h](#)
- src/[student.cpp](#)

Chapter 5

File Documentation

5.1 Include/human.h File Reference

```
#include "my_library.h"
```

Classes

- class [Human](#)

5.2 human.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "my_library.h"
00004
00005 class Human {
00006     protected:
00007         string vardas;
00008         string pavarde;
00009
00010     public:
00011     Human(const string& vardas, const string& pavarde)
00012         : vardas(vardas), pavarde(pavarde)
00013     { }
00014
00015     string getVardas() const { return vardas; }
00016     string getPavarde() const { return pavarde; }
00017
00018     virtual void unusedFunction() const = 0;
00019
00020     // Virtual destructor for proper cleanup
00021     virtual ~Human() {
00022         vardas.clear();
00023         pavarde.clear();
00024     }
00025
00026 };
```

5.3 Include/my_functions.h File Reference

```
#include "my_library.h"
#include "student.h"
```

Functions

File Utilities (file_functions.cpp)

- `std::string create_folder` (const `std::string` &folder_name)
Create a directory if it does not exist.
- `void generate_files` (int student_size)
Generate test files with random student records.

Student Management (student_functions.cpp)

- `void print_students` (const `std::vector< Studentas >` &students)
Print a header and list of students to console.
- `void insert_student` (`std::vector< Studentas >` &student_list, `Studentas` &student)
Append a student object to the list via move semantics.
- `void sort_students` (`std::vector< Studentas >` &student_list)
Sort students by user-selected criterion.
- `void sort_students` (`std::vector< Studentas >` &student_list, int choice)
Sort students by a programmatic choice.

User Choices (program_choice.cpp)

- `Studentas userChoice_2` ()
Prompt user for manual input with random grade generation (choice 2).
- `Studentas userChoice_3` ()
Generate both name and grades randomly (choice 3).

Random Data Generation (random_generating.cpp)

- `int randomNumber` (int a, int b)
Produce a uniformly distributed integer within [a, b].
- `std::vector< float > random_grade` ()
Generate a vector of 2–8 random float grades (1–10).
- `std::string random_name` ()
Select a random first name from a fixed list.
- `std::string random_last_name` ()
Select a random surname from a fixed list.

Input Validation (additional_functions.cpp)

- `int check_the_value` (const `std::string` &message, const `std::string` &errorMessage, int minVal, int maxVal)
Prompt for an integer within a range, re-prompting on invalid input.
- `std::string consoleText_userChoice` ()
Display the main menu and return user choice string.

Testing Routines (testavimas.cpp)

- `void testing_v12` ()
Interactive tests for constructors, assignment, and I/O.
- `void testing_v11` ()
Batch performance tests for file reading, sorting, and grouping.
- `void strategy_1` ()
Strategy 1: read, sort, divide via unoptimized loops, and write results.
- `void strategy_2` ()
Strategy 2: read, sort, divide via pop_back, and measure times.
- `void strategy_3` ()
Strategy 3: read, sort, divide via find/erase, and measure times.

5.3.1 Function Documentation

5.3.1.1 `check_the_value()`

```
int check_the_value (
    const std::string & message,
    const std::string & errorMessage,
    int minVal,
    int maxVal)
```

Prompt for an integer within a range, re-prompting on invalid input.

Parameters

<i>message</i>	Prompt displayed to user.
<i>errorMessage</i>	Error text for out-of-range or non-integer input.
<i>minVal</i>	Minimum valid value.
<i>maxVal</i>	Maximum valid value.

Returns

Validated integer in [minVal, maxVal].

5.3.1.2 `consoleText_userChoice()`

```
std::string consoleText_userChoice ()
```

Display the main menu and return user choice string.

Returns

Raw menu input string for numeric parsing.

5.3.1.3 `create_folder()`

```
std::string create_folder (
    const std::string & folder_name)
```

Create a directory if it does not exist.

Parameters

<i>folder_name</i>	Name of the folder to create (relative to CWD).
--------------------	---

Returns

Full path to the created or existing folder, or empty string on failure.

Note

On failure to create, prints an error to std::cerr.

5.3.1.4 `generate_files()`

```
void generate_files (
    int student_size)
```

Generate test files with random student records.

Parameters

<i>student_size</i>	Exponent used to compute record count: $10^{\text{student_size}}$.
---------------------	--

Creates a "Generated Files" folder, then writes a header (name, surname, 5 NDs, exam) and $\langle \text{size} \rangle$ lines of randomly generated data into "testavimasFailas $\langle \text{size} \rangle$.txt". Upon completion, prints the file creation message and path to `std::cout`.

5.3.1.5 insert_student()

```
void insert_student (
    std::vector< Studentas > & student_list,
    Studentas & student)
```

Append a student object to the list via move semantics.

Parameters

<i>student_list</i>	Vector to which the student is added.
<i>student</i>	<code>Studentas</code> instance to insert (moved).

5.3.1.6 print_students()

```
void print_students (
    const std::vector< Studentas > & students)
```

Print a header and list of students to console.

Parameters

<i>students</i>	Vector of <code>Studentas</code> to display.
-----------------	--

Uses fixed-width columns to show surname, name, mean- and median-based final grades.

5.3.1.7 random_grade()

```
std::vector< float > random_grade ()
```

Generate a vector of 2–8 random float grades (1–10).

Returns

Vector with randomly determined length and grade values.

5.3.1.8 random_last_name()

```
std::string random_last_name ()
```

Select a random surname from a fixed list.

Returns

Random surname string.

5.3.1.9 random_name()

```
std::string random_name ()
```

Select a random first name from a fixed list.

Returns

Random name string.

5.3.1.10 randomNumber()

```
int randomNumber (
    int a,
    int b)
```

Produce a uniformly distributed integer within [a, b].

Parameters

<i>a</i>	Lower bound (inclusive).
<i>b</i>	Upper bound (inclusive).

Returns

Random integer between a and b.

5.3.1.11 sort_students() [1/2]

```
void sort_students (
    std::vector< Studentas > & student_list)
```

Sort students by user-selected criterion.

Parameters

<i>student_list</i>	Vector of Studentas to sort.
---------------------	--

Prompts the user with four options: name, surname, mean, or median.

5.3.1.12 sort_students() [2/2]

```
void sort_students (
    std::vector< Studentas > & student_list,
    int choice)
```

Sort students by a programmatic choice.

Parameters

<i>student_list</i>	Vector of Studentas to sort.
<i>choice</i>	Criterion: 1=name, 2=surname, 3=mean, 4=median.

5.3.1.13 strategy_1()

```
void strategy_1 ()
```

Strategy 1: read, sort, divide via unoptimized loops, and write results.

<— sparta pradedama skaiciuoti cia

5.3.1.14 strategy_2()

```
void strategy_2 ()
```

Strategy 2: read, sort, divide via `pop_back`, and measure times.

5.3.1.15 strategy_3()

```
void strategy_3 ()
```

Strategy 3: read, sort, divide via `find/erase`, and measure times.

5.3.1.16 testing_v11()

```
void testing_v11 ()
```

Batch performance tests for file reading, sorting, and grouping.

5.3.1.17 testing_v12()

```
void testing_v12 ()
```

Interactive tests for constructors, assignment, and I/O.

5.3.1.18 userChoice_2()

```
Studentas userChoice_2 ()
```

Prompt user for manual input with random grade generation (choice 2).

Returns

[Studentas](#) initialized with entered name/surname and random grades/exam.

5.3.1.19 userChoice_3()

```
Studentas userChoice_3 ()
```

Generate both name and grades randomly (choice 3).

Returns

`Studentas` initialized with random name, surname, grades, and exam.

5.4 my_functions.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "my_library.h"
00004 #include "student.h"
00005
00009
00010 // -----
00013
00018 std::string create_folder(const std::string& folder_name);
00019
00025 void generate_files(int student_size);
00026
00028
00029 // -----
00032
00036 void print_students(const std::vector<Studentas>& students);
00037
00041 void insert_student(std::vector<Studentas>& student_list, Studentas& student);
00042
00046 void sort_students(std::vector<Studentas>& student_list);
00047
00051 void sort_students(std::vector<Studentas>& student_list, int choice);
00052
00054
00055 // -----
00058
00061 Studentas userChoice_2();
00062
00065 Studentas userChoice_3();
00066
00068
00069 // -----
00072
00077 int randomNumber(int a, int b);
00078
00081 std::vector<float> random_grade();
00082
00085 std::string random_name();
00086
00089 std::string random_last_name();
00090
00092
00093 // -----
00096
00103 int check_the_value(const std::string& message,
00104                     const std::string& errorMessage,
00105                     int minVal,
00106                     int maxVal);
00107
00110 std::string consoleText_userChoice();
00111
00113
00114 // -----
00117
00119 void testing_v12();
00120
00122 void testing_v11();
00123
00125 void strategy_1();
00126
00128 void strategy_2();
00129
00131 void strategy_3();
00132
```

5.5 Include/my_library.h File Reference

```
#include <vector>
#include <iostream>
#include <fstream>
#include <sstream>
#include <string>
#include <iomanip>
#include <algorithm>
#include <iterator>
#include <limits>
#include <numeric>
#include <random>
#include <cmath>
#include <chrono>
#include <filesystem>
```

5.6 my_library.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003
00004
00005 // --- Containers ---
00006 #include <vector>
00007 using std::vector;
00008
00009 // --- I/O Streams ---
00010 #include <iostream>
00011 using std::cout;
00012 using std::cin;
00013 using std::cerr;
00014 using std::endl;
00015 using std::ostream;
00016 using std::ios;
00017
00018 #include <fstream>
00019 using std::ifstream;
00020 using std::ofstream;
00021
00022 #include <sstream>
00023 using std::stringstream;
00024 using std::ostringstream;
00025
00026 // --- String and Formatting ---
00027 #include <string>
00028 using std::string;
00029 using std::to_string;
00030
00031 #include <iomanip>
00032 using std::setw;
00033 using std::fixed;
00034 using std::setprecision;
00035 using std::left;
00036
00037 // --- Algorithms and Iterators ---
00038 #include <algorithm>
00039 using std::sort;
00040
00041 #include <iterator>
00042 using std::istream_iterator;
00043
00044 // --- Numeric Utilities ---
00045 #include <limits>
00046 using std::numeric_limits;
00047 using std::streamsize;
00048
00049 #include <numeric>
00050 using std::accumulate;
```

```

00051
00052 // --- Random Number Generation ---
00053 #include <random>
00054 using std::random_device;
00055 using std::mt19937;
00056 using std::uniform_int_distribution;
00057
00058 // --- Math ---
00059 #include <cmath>
00060
00061 // --- Time Utilities ---
00062 #include <chrono>
00063 using std::chrono::high_resolution_clock;
00064 using std::chrono::duration;
00065
00066 #include <filesystem>

```

5.7 Include/student.h File Reference

Declaration of the [Studentas](#) class, extending [Human](#) to manage grades and compute final scores.

```

#include "human.h"
#include "my_library.h"

```

Classes

- class [Studentas](#)

Represents a student with homework grades and an exam score, providing calculations for mean- and median-based final grades.

5.7.1 Detailed Description

Declaration of the [Studentas](#) class, extending [Human](#) to manage grades and compute final scores.

5.8 student.h

[Go to the documentation of this file.](#)

```

00001 #pragma once
00002
00003 #include "human.h"
00004 #include "my_library.h"
00005
00008
00012 class Studentas : public Human {
00013 public:
00015     Studentas()
00016         : Human("", ""), egzaminoRezultatas(0),
00017           galutinisVid(0.0f), galutinisMed(0.0f) {}
00018
00028     Studentas(const std::string& vardas,
00029               const std::string& pavarde,
00030               const std::vector<float>& pazymiai,
00031               const int& egzaminoRezultatas)
00032         : Human(vardas, pavarde), pazymiai(pazymiai), egzaminoRezultatas(egzaminoRezultatas) {
00033         calculate_everything();
00034     }
00035
00037     ~Studentas() {
00038         vardas.clear();
00039         pavarde.clear();
00040         pazymiai.clear();
00041         egzaminoRezultatas = 0;

```

```

00042         galutinisVid = 0.0f;
00043         galutinisMed = 0.0f;
00044     }
00045
00046     Studentas(const Studentas& other);
00047
00048     Studentas(Studentas&& other) noexcept;
00049
00050     Studentas& operator=(const Studentas& other);
00051
00052     Studentas& operator=(Studentas&& other) noexcept;
00053
00054     // Setters
00055     void setVardas(const std::string& v) { vardas = v; }
00056     void setPavarde(const std::string& p) { pavarde = p; }
00057     void setGrades(const std::vector<float>& g) { pazymiai = g; }
00058     void setExam(const int& exam) { egzaminoRezultatas = exam; }
00059
00060     // Getters
00061     std::vector<float> getPazymiai() const { return pazymiai; }
00062     int getEgzaminoRezultatas() const { return egzaminoRezultatas; }
00063     float getGalutinisVid() const { return galutinisVid; }
00064     float getGalutinisMed() const { return galutinisMed; }
00065
00066     void unusedFunction() const override;
00067
00068     float calculateMean();
00069
00070     float calculateMedian();
00071
00072     void calculate_everything();
00073
00074     friend std::ostream& operator<<(std::ostream& out, const Studentas& s);
00075
00076     friend std::istream& operator>>(std::istream& in, Studentas& s);
00077
00078     static void appendingContainerViaFile(const std::string& file_name,
00079                                           std::vector<Studentas>& student_list);
00080
00081     static std::vector<Studentas> read_student_records(int ndCount,
00082                                                         std::istream& iss);
00083
00084     static void print_to_file(const std::vector<Studentas>& list_of_students,
00085                               const std::string& file_name);
00086
00087 private:
00088     std::vector<float> pazymiai;
00089     int egzaminoRezultatas;
00090     float galutinisVid;
00091     float galutinisMed;
00092 };

```

5.9 src/additional_functions.cpp File Reference

```

#include "my_library.h"
#include "my_functions.h"

```

Functions

- int [check_the_value](#) (string message, string errorMessage, int minVal, int maxVal)
- string [consoleText_userChoice](#) ()

Display the main menu and return user choice string.

5.9.1 Function Documentation

5.9.1.1 check_the_value()

```

int check_the_value (
    string message,
    string errorMessage,
    int minVal,
    int maxVal)

```

5.9.1.2 consoleText_userChoice()

```
string consoleText_userChoice ()
```

Display the main menu and return user choice string.

Returns

Raw menu input string for numeric parsing.

5.10 src/file_functions.cpp File Reference

```
#include "my_library.h"
#include "my_functions.h"
```

Functions

- string [create_folder](#) (string folder_name)
- void [generate_files](#) (int student_size)
Generate test files with random student records.

5.10.1 Function Documentation

5.10.1.1 create_folder()

```
string create_folder (
    string folder_name)
```

5.10.1.2 generate_files()

```
void generate_files (
    int student_size)
```

Generate test files with random student records.

Parameters

<i>student_size</i>	Exponent used to compute record count: $10^{\text{student_size}}$.
---------------------	--

Creates a "Generated Files" folder, then writes a header (name, surname, 5 NDs, exam) and <size> lines of randomly generated data into "testavimasFailas<size>.txt". Upon completion, prints the file creation message and path to std::cout.

5.11 src/main.cpp File Reference

```
#include "my_library.h"  
#include "my_functions.h"
```

Functions

- int [main](#) ()

5.11.1 Function Documentation

5.11.1.1 main()

```
int main ()
```

5.12 src/program_choice.cpp File Reference

```
#include "my_library.h"  
#include "my_functions.h"
```

Functions

- [Studentas userChoice_2](#) ()
Prompt user for manual input with random grade generation (choice 2).
- [Studentas userChoice_3](#) ()
Generate both name and grades randomly (choice 3).

5.12.1 Function Documentation

5.12.1.1 userChoice_2()

```
Studentas userChoice_2 ()
```

Prompt user for manual input with random grade generation (choice 2).

Returns

[Studentas](#) initialized with entered name/surname and random grades/exam.

5.12.1.2 userChoice_3()

```
Studentas userChoice_3 ()
```

Generate both name and grades randomly (choice 3).

Returns

`Studentas` initialized with random name, surname, grades, and exam.

5.13 src/random_generating.cpp File Reference

```
#include "my_library.h"  
#include "my_functions.h"
```

Functions

- int `randomNumber` (int a, int b)
Produce a uniformly distributed integer within [a, b].
- vector< float > `random_grade` ()
Generate a vector of 2–8 random float grades (1–10).
- string `random_name` ()
Select a random first name from a fixed list.
- string `random_last_name` ()
Select a random surname from a fixed list.

5.13.1 Function Documentation

5.13.1.1 random_grade()

```
vector< float > random_grade ()
```

Generate a vector of 2–8 random float grades (1–10).

Returns

Vector with randomly determined length and grade values.

5.13.1.2 random_last_name()

```
string random_last_name ()
```

Select a random surname from a fixed list.

Returns

Random surname string.

5.13.1.3 random_name()

```
string random_name ()
```

Select a random first name from a fixed list.

Returns

Random name string.

5.13.1.4 randomNumber()

```
int randomNumber (
    int a,
    int b)
```

Produce a uniformly distributed integer within [a, b].

Parameters

<i>a</i>	Lower bound (inclusive).
<i>b</i>	Upper bound (inclusive).

Returns

Random integer between a and b.

5.14 src/student.cpp File Reference

```
#include "my_library.h"
#include "student.h"
#include "my_functions.h"
```

Functions

- ostream & [operator<<](#) (ostream &out, const [Studentas](#) &s)
- std::istream & [operator>>](#) (std::istream &in, [Studentas](#) &s)

5.14.1 Function Documentation

5.14.1.1 operator<<()

```
ostream & operator<< (
    ostream & out,
    const Studentas & s)
```

5.14.1.2 operator>>()

```
std::istream & operator>> (  
    std::istream & in,  
    Studentas & s)
```

5.15 src/student_functions.cpp File Reference

```
#include "my_library.h"  
#include "my_functions.h"
```

Functions

- void `print_students` (vector< `Studentas` > &studentas)
- void `insert_student` (vector< `Studentas` > &student_list, `Studentas` &student)
- void `sort_students` (vector< `Studentas` > &student_list)
- void `sort_students` (vector< `Studentas` > &student_list, int choice)

5.15.1 Function Documentation

5.15.1.1 insert_student()

```
void insert_student (  
    vector< Studentas > & student_list,  
    Studentas & student)
```

5.15.1.2 print_students()

```
void print_students (  
    vector< Studentas > & studentas)
```

5.15.1.3 sort_students() [1/2]

```
void sort_students (  
    vector< Studentas > & student_list)
```

5.15.1.4 sort_students() [2/2]

```
void sort_students (  
    vector< Studentas > & student_list,  
    int choice)
```

5.16 src/testavimas.cpp File Reference

```
#include "my_library.h"
#include "my_functions.h"
```

Functions

- string [consoleText_testing12](#) ()
- template<typename Func>
double [measureTime](#) (Func func)
- void [testing_v12](#) ()
Interactive tests for constructors, assignment, and I/O.
- void [testing_v11](#) ()
Batch performance tests for file reading, sorting, and grouping.
- vector< [Studentas](#) > [divide_students_unoptimized](#) (const vector< [Studentas](#) > &list_of_students, int choice)
- void [strategy_1](#) ()
Strategy 1: read, sort, divide via unoptimized loops, and write results.
- void [strategy_2](#) ()
Strategy 2: read, sort, divide via pop_back, and measure times.
- void [strategy_3](#) ()
Strategy 3: read, sort, divide via find/erase, and measure times.

5.16.1 Function Documentation

5.16.1.1 consoleText_testing12()

```
string consoleText_testing12 ()
```

5.16.1.2 divide_students_unoptimized()

```
vector< Studentas > divide_students_unoptimized (  
    const vector< Studentas > & list_of_students,  
    int choice)
```

5.16.1.3 measureTime()

```
template<typename Func>  
double measureTime (  
    Func func)
```

5.16.1.4 strategy_1()

```
void strategy_1 ()
```

Strategy 1: read, sort, divide via unoptimized loops, and write results.

<— sparta pradedama skaiciuoti cia

5.16.1.5 strategy_2()

```
void strategy_2 ()
```

Strategy 2: read, sort, divide via pop_back, and measure times.

5.16.1.6 strategy_3()

```
void strategy_3 ()
```

Strategy 3: read, sort, divide via find/erase, and measure times.

5.16.1.7 testing_v11()

```
void testing_v11 ()
```

Batch performance tests for file reading, sorting, and grouping.

5.16.1.8 testing_v12()

```
void testing_v12 ()
```

Interactive tests for constructors, assignment, and I/O.

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